# KUSAL YAPA

Trainee Software Engineer | Final Year Student BEng (Hons) Software Engineering


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## Profile

Dynamic undergraduate Software Engineering student at the University of Westminster, graduating in September 2026, currently honing skills as a trainee Software Engineer at Horizon Group USA. Experience at 99x provided invaluable hands-on expertise in full-stack development, agile methodologies, and effective problem-solving, fostering adaptability to emerging technologies. A keen interest in AI, Machine Learning, and scalable software systems drives a commitment to continuous learning and the ambition to create impactful solutions. Equipped with a strong foundation in programming languages such as Java, Python, and TypeScript dedication to leveraging technical expertise and creativity ensures

## Skills

| | |
|---|---|
| Programming Languages | Java, JavaScript, TypeScript, Python, Swift |
| Frameworks & Libraries | Next.js, Nest.js, React.js, Vue.js |
| Tools & Platforms | GitHub, AWS, Azure project management, Figma, VSCode |
| Data management | PostgreSQL, MongoDB, AWS RDS |
| Soft Skills | Team player, Communication, Adaptability, Continuous Learning |

## Work Experience

- **Trainee Software Engineer -** *Horizon Group USA, Colombo, Sri Lanka* 10/2025 – Present

Demand-Planning-Tool: An enterprise web-based Demand Planning application to replace legacy planning tools. Implemented role-based authentication using Active Directory and built modules for product allocation, monthly forecast planning, bulk forecast updates, and demand normalization. Integrated Data Warehouse data for historical sales, inventory metrics, and forecasting analysis. Developed threshold-based exception handling with reason tracking and audit logging. Ensured data validation, scalable system design, and improved forecast accuracy and planning efficiency.

- **Trainee Software Engineer -** *99x Technology Limited, Colombo, Sri Lanka* 07/2023 – 07/2024

Employee Upskilling and Reskilling Application: Contributed as a team member, designing and developing an internal platform for employee skill development. My role involved ensuring seamless backend integration and creating a user-friendly experience that enables employees to customize their learning journeys.

AI Web Solutions Platform: Contributed to the development of a modern web application, utilizing Vue.js, and Nuxt.js. The application focused on form submissions, real-time validation, SEO optimization, and external platform integration. My role involved designing and optimizing the system to ensure scalability, user engagement, and visibility.

AI - HR Bot: Worked on building an AI-powered HR Bot solution, integrating OpenAI for actionable insights and using vector databases for efficient data retrieval. My role involved enhancing decision-making through prompt engineering and streamlining HR tasks to improve overall efficiency.

AI Research Hub: Contributed to the development of a platform showcasing AI research and case studies using Next.js and Vue.js. My role involved creating an intuitive and informative interface to highlight AI advancements and the company's global contributions.

Optimaxer AI Platform: Contribute to the development of a platform with Agent Creation and the extraction tools with the AI-based Chat Bot, and the product award winners as a team, contributed to the OPTIMAXER AI project, NBQSA Merit Award Winner (2023), corporate research and development.

## Education

**University of Westminster collaborate Informatics Institute of Technology – IIT**
- BEng (Hons) Software Engineering | 2021 – 2026

**ESOFT Metro Campus**
- Diploma in Information Technology | 2020 – 2021

**Richmond College**
- Biology, Chemistry, Physics | 2006 – 2019

## Modules and Marks - BEng (Hons) Software Engineering

**1st Year,** *Avg- 85.1%*

- Software Development I
- Mathematics for Computing
- Trends in Computer Science
- Computer System Fundamentals
- Software Development II
- Web Design and Development

**2nd Year,** *Avg - 71.66%*

- Object Oriented Programming
- Database Systems
- Software Development Group Project
- Mobile Application Development
- Machine Learning and Data Mining
- Algorithms: Theory, Design and Implementation

**4th Year,** *Avg - 59.2% (one subject pending)*

- Mobile Native App Development
- Computer Science Final Project
- Formal Methods
- Concurrent Programming
- Cyber Security (pending)

**3rd Year,** *Industrial Placement at 99x*

## Achievements and Certifications

2024-2025

- AWS Educate: Introduction to Generative AI
- AWS Educate Machine Learning Foundations
- AWS Educate Introduction to Cloud 101

2022-2023

- NBQSA Merit Award Winner (2023) corporate research and development. Marit Awards winners as a team
- Contributed to the OPTIMAXER AI project

2021-2022

- OpenHack 2.0 Designathon 1st Runner up
- Design a prototype for a home energy management system using modern UI/UX concepts.
- Finalist team member of RevolUX 2.0 UI/UX
- Designathon organized by AIESEC of the University of Colombo School of computing
- Participated in IEEEXtreme 16.0 Programming - Island 67 Organized by IEEE at IIT
- Participant to hacktoberfest 2022

## Projects

**Final Year Project - FYP - Adaptive Gamification Model for Enhancing Programming Education**

For my final year project, I designed and implemented an adaptive gamification model aimed at enhancing programming education. This system utilized reinforcement learning (DQN) and basic gamification elements like feedback, Progress Bar, leaderboards, and real-time feedback to personalize and adapt quizzes based on the learner's knowledge level and pace. The system dynamically adjusted the difficulty of programming tasks, offering personalized feedback to keep students engaged and motivated, reducing dropout rates, and improving knowledge retention. The learning experience for Gen Z students, is normalized to game-like environments. The project aimed to bridge gaps in traditional programming education by making learning more engaging, interactive, and tailored to each student's needs.

**SDGP - Software development Group project**

TruAttend is an online virtual meeting platform that utilizes facial recognition and mobile phone detection for secure access and accurate attendance tracking. The platform features effective monitoring and comprehensive reporting to improve the quality of lectures and enhance the online learning experience for students.

## Contribution To Cross-Functional initiatives and Volunteer Activities

**Member**

2023 – 2024 Music Innovative at 99X

**Volunteer**

2023 – 2024 Volunteered at Xternship program held at IIT, Sri Lanka, for BIS students in web development

2022 – 2023 Volunteered at STEM UP, which is a foundation that provides support to students who are inspired to be programmers.

## More About Me

Visit— https://linktr.ee/kusal_madura_yapa